Minimum Coin Change

Topic: DynamicProgramming File: dp4.pdf

PROBLEM

Given coin denominations and an amount A, find the minimum number of coins needed to make A, or report that it is impossible.

STATE

dp[x] stores the minimum coins needed to make amount x. Set dp[0] = 0 and other states initially to infinity.

TRANSITION

For each amount x and each coin c <= x, update dp[x] = min(dp[x], dp[x-c] + 1) if dp[x-c] is reachable.

EXAMPLE

Coins [1,3,4], amount 6. The minimum is 2 coins: 3 + 3. A greedy choice of 4 first would need 4 + 1 + 1.

COMPLEXITY

For A as the target amount and k coin types: O(Ak) time and O(A) space.

IMPORTANT

The dynamic-programming method works for arbitrary positive denominations; greedy change-making does not always give the optimum.